Comparative Evaluation of Machine Learning Models for Credit Scoring in FinTech Applications

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***Abstract*—Proper credit scoring is also crucial for controlling financial risks and targeting customers in the financial technology (FinTech) industry. Although conventional statistical modeling has always been the modeling choice, its successor, modern machine learning (ML) algorithms, demonstrates more outstanding predictive power. However, these models often involve a trade-off between predictive accuracy and interpretability. This study contrasts the results of three well-known ML models: Logistic Regression, Random Forest, and XGBoost. The tools were applied to a real-life dataset of a financial institution, namely, personal loan data. The evaluation metrics used were accuracy, precision, recall, F1-score, and Area Under the Curve (AUC), receiver operating characteristic. According to the results, XGBoost achieves the highest overall performance, with an impressive accuracy of 98.5% and a recall score of 96.9%. Close behind was Random Forest with its competitive performance. Logistic Regression provided better interpretability, but it was not as accurate as recall. This weak point suggests that more people who could actually be creditworthy will be discounted as applicants. These insights enable practitioners to make informed decisions about the most suitable ML methods to adopt in credit scoring tasks, particularly where accuracy and sensitivity are key considerations. In addition, the current analysis will establish a performance reference that will be useful when attempting to enhance transparency and fairness in future credit decision processes driven by AI.**

***Index Terms*—FinTech, Credit Scoring, Machine Learning, XGBoost, Transparency, Model Interpretability, Regulatory Compliance**

I. Introduction

The accelerated digital transformation in the financial sector has led to the widespread adoption of artificial intelligence (AI) and machine learning (ML) techniques, particularly for automating complex decision-making processes [1]. One prominent application of these technologies is credit scoring, a data-driven approach used to assess the creditworthiness of loan applicants. AI-powered robo-advisors are reshaping the financial advisory landscape by leveraging data-driven algorithms to deliver personalized portfolio recommendations with minimal human intervention [2]. They also streamline customer service and decision-making processes through the use of contextual intelligence [3], [4]. Unlike traditional rule-based systems, ML models offer substantial improvements in efficiency and cost-effectiveness for financial institutions [5]. Notably, these algorithms are well-suited to capture intricate, non-linear patterns in borrower data and are capable of identifying subtle risk indicators that conventional methods might overlook. Although ML models have good predictive capabilities, they are a challenge in real-life financial applications. A serious issue is the trade-off between predictive accuracy and interpretability. Learning-based ensemble models, such as Random Forest and XGBoost, exhibit very high predictive performance but are widely regarded as opaque or black-box models [6]. This lack of transparency raises serious questions regarding transparency, auditability, and adherence to regulatory standards, such as the General Data Protection Regulation (GDPR) and the Equal Credit Opportunity Act (ECOA) [7]. In comparison, less complex models, such as Logistic Regression, afford a deeper view into their results at the expense of not performing as well as more sophisticated approaches, especially when attempting to use classification models with imbalanced datasets. In this light, the systematic benchmarking of ML models, regarding not only accuracy but also their applicability in real-life financial contexts, is becoming increasingly essential. This is particularly crucial in credit scoring activities, where mistakes can have serious financial and reputational consequences. Although many studies have been conducted previously, focusing solely on maximizing predictive performance or investigating the interpretability of the model, none have conducted a comprehensive comparative study based on several major evaluation measures using practical financial data. In this study, the identified gap was filled by performing a comparative analysis of the three most commonly used ML algorithms, including Logistic Regression, Random Forest, and XGBoost, through a real-life situation of a personal loan provided by a bank [8]. This aims to assess how well each model addresses class imbalance, its predictive accuracy, and how effectively it selects and interprets features. Although explainable AI (XAI) techniques are not included in this study, they establish a necessary benchmark for future studies in the same direction. The results of this study will help FinTech professionals and financial decision-makers comprehend the trade-offs that are essential when choosing different models for credit risk assessment.

II. Literature Review

Credit scoring has long been a common element of both consumer and business lending, with much of it resting on statistical modeling methods such as logistic regression and linear

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discriminant analysis [9]. Traditional methods are interpretable and easily applicable, making them suitable for regulated financial aspects. Recent research highlights the effectiveness of ensemble AutoML approaches in accurately predicting STEM student grades using multi-class classification models [10]. However, with the increased complexity and volume of financial data, traditional approaches are becoming less effective in modeling the complex, non-linear trends and interdependencies of variables captured through financial data [11], resulting in inaccurate assessments of credit risk. Over the past few years, the financial technology (FinTech) sector has been using machine learning (ML) approaches to advance credit scoring models [12]. Numerous algorithms, including decision forests, random forests, support vector machines (SVMs), and gradient boosting techniques, specifically XGBoost [13], were tested to improve predictive accuracy. As indicated in the benchmarking study by Lessmann et al., ensemble techniques, in general, and boosting techniques [14], in particular, consistently demonstrate substantial improvements over conventional models across a wide range of datasets. In the same breath, Bahnsen et al. emphasized the usefulness of cost-sensitive learning in minimizing misclassification costs in credit scoring cases [15]. Random Forest and XGBoost are among these methods, and both have generated significant interest in industry and academia due to their robustness against overfitting and versatility in handling mixed feature types [16]. Specifically, XGBoost takes additional steps to enhance predictive performance by incorporating built-in regularization terms, tree pruning, and parallel training aspects [17], which make it highly successful in applications involving high-dimensional and imbalanced datasets. However, despite their high performance, such ensemble models are often criticized for their minimal interpretability. Transparency, auditability, and compliance with regulations, including the General Data Protection Regulation (GDPR) [18] and the Equal Credit Opportunity Act (ECOA) [19], are of particular concern in high-stakes financial applications. Comparative studies on credit risk modelling consistently demonstrate that critical trade-offs affect the decision between the price model selection. For example, Brown and Mues observed that, whereas logistic regression remains a good base, ensemble solutions tend to score higher in recall and AUC-ROC [20], albeit at the expense of greater complexity and a more challenging regulatory environment. In a similar vein, Abdo [21] highlighted the critical need to tune and validate a model to utilize advanced ML methods and achieve higher predictive accuracy. Although more studies have emerged in the field of explainability methods, the present study aimed to establish a solid performance baseline for simple ML methods [22], thereby enabling further studies on the interpretability of models. This study helps answer the question of how to choose between logistic regression and random forest or XGBoost by providing a comprehensive comparison of these methods in a real example of financial data. It can serve as a valuable source of information when deciding which of these models to use in cases where there is a class imbalance in the data, high regulatory requirements, and a high degree of regulatory oversight.

III. Methodology

A. Dataset Description and Preprocessing: This study utilizes a publicly available dataset from a retail bank, comprising 1,006 customer records related to personal loan applications [23]. Each record includes demographic and financial attributes such as age, income, family size, credit card ownership, education level, account balances, and a binary indicator representing the loan approval status. The dataset exhibits a significant class imbalance, with only 9.6% of the applicants receiving a loan, making it particularly suitable for evaluating the sensitivity of predictions for the minority class. A thorough preprocessing pipeline was applied before the model was developed. Missing values were checked, although the dataset was largely complete. Categorical variables, such as education and family size, were transformed using one-hot encoding to facilitate their integration with ML algorithms. Numeric features were standardized using Z-score normalization when necessary. The dataset was split into training and testing sets, with 70% allocated to training and 30% to testing, using stratified sampling to maintain a consistent distribution of the target class. B. Model Selection and Training Three ML classification models, Logistic Regression, Random Forest, and XGBoost, were selected for comparative analysis. Logistic Regression served as a baseline model owing to its interpretability and widespread use in credit-scoring applications. Random Forest and XGBoost were selected for their proven strengths in identifying complex, nonlinear relationships and enhancing predictive performance in structured financial data. All models were implemented in Python using the Scikit-learn and XGBoost libraries. Hyperparameter tuning was conducted using a grid search combined with 5-fold cross-validation on the training set. The tuned parameters included the number of estimators and maximum depth for Random Forest, as well as learning rate, tree depth, and subsampling ratios for XGBoost. Logistic Regression was trained using $L2$ regularization to reduce the risk of overfitting. C. Evaluation Metrics: Several performance metrics were employed to evaluate the models comprehensively. These included accuracy, precision, recall, $F1$-score, and area under the ROC curve (AUC-ROC). While accuracy provides a general measure of correctness, precision and recall offer detailed insights into the effectiveness of the predictions for the minority class. The $F1$-score balances these two metrics, representing the harmonic mean of precision and recall. The AUC-ROC was particularly helpful in assessing the discriminative ability of the models between loan approvals and rejections across various thresholds. Confusion matrices were generated to visually assess the true positives, false positives, and false negatives. Given the class imbalance, particular emphasis was placed on recall to identify creditworthy customers accurately. The experiments were conducted in a local computing environment using Python 3.11 within a Jupyter Notebook. Essential libraries, including Scikit-learn, XGBoost, Pandas, NumPy, and Matplotlib, were utilized for data manipulation, model building, and visualization.

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IV. Results

The Fig. 1 shows that the best baseline was Logistic Regression, with an accuracy of 95.4%, precision of 82.1%, recall rate of 66.7%, and $F1$-score of 73.6. It achieved an AUC-ROC of 96.4%, indicating good discriminative power. Nevertheless, although it can be easily interpreted and provides consistent results, logistic regression cannot strongly capture the minority group. The confusion matrix indicates that it detected 64 of 96 actual loan approvals but missed 32, resulting in false negatives, as shown in the matrix. It was very accurate, but its conservative prediction model did not allow it to find credit-worthy applicants entirely.

This was because the Random Forest model remarkably outperformed Logistic Regression, which demonstrated insensitivity to the positive class. It also achieved a very good accuracy of 98.2%, precision of 87.5%, recall of 94.8%, and $F1$-score of 91.0. In addition, the AUC-ROC score of 99.8 presented in the model indicated that the model had very high performance in distinguishing between approved and non-approved loan cases. By examining the confusion matrix, only five false negatives and 13 false positives were recorded, indicating its good predictive power and ability to deal with unbalanced datasets effectively. However, in this assessment, the XGBoost model showed the best results for all essential indicators. It achieved an accuracy of 98.5 percent, precision of 88.6%, recall of 96.9%, and an $F1$-score of 92.5%, with an AUC-ROC of 99.9%, which was nearly ideal. Notably, the model provided the wrong classification for only three approved loans and 12 non-approved loans, which signifies the high effectiveness of the model in correctly identifying credit-worthy applicants without a substantial increase in false positives. The precision-recall graph also confirmed the excellent performance of XGBoost in addressing the imbalance problem, with significant average precision and consistency across multiple levels of precision and recall. The analysis of the feature importance of all the analyzed models revealed that Income, Education, CD Account ownership, and Credit Card possession are the most significant factors in predicting loan approval. Although Logistic Regression provided easy-to-interpret and clear coefficients, the ensemble models, such as Random Forest and XGBoost, revealed more about the complicated and nonlinear causalities between features, albeit at the cost of interpretability. In summary, it was made clear that the ensemble-based algorithms, as a whole, especially Random Forest and XGBoost, outperformed the Logistic Regression algorithm in all facets of important performance measures and excelled most in recall, which is of crucial importance in ensuring the best possible false-negative rates of credit decision systems. The end-to-end predictive accuracy and reliability of XGBoost are advantageous, especially in the context of FinTech, where accuracy and sensitivity in distinguishing among applicants willing to be served with loans are crucial. These can serve as a solid empirical basis to inform the selection of a model for real-world credit scoring applications.

V. Conclusion and Future Work

This study conducts a comparative analysis of three machine learning algorithms: Logistic Regression, Random Forest, and XGBoost, using real-world banking data to predict personal loan approval outcomes. The results indicate that, despite being a powerful and simple-to-interpret baseline, logistic regression is significantly outperformed by the ensemble-based models in terms of predictive accuracy [24], sensitivity to the minority class, and general classification performance. Among the considered models, XGBoost was the most successful in delivering the best outcomes for all key metrics. In particular, it achieved an accuracy of 98.5% and a recall of 96.9%, indicating that it is skilled at accurately predicting which applicants will be approved for loans. The random Forest was also quite good and balanced well in terms of precision and recall [25]. However, due to its relative accuracy, logistic regression has a relatively low recall value, implying a high potential for missing appropriate candidates, which is a critical disadvantage in credit scoring. These findings have critical implications for practitioners in FinTech and financial organizations interested in adopting AI-based credit scoring models. The proven advantage of ensemble methods in terms of performance demonstrates that they are well-suited for operational conditions where risk prediction and management are required, and accurate predictions are necessary. However, this improved predictive performance is counterbalanced by lower interpretability, which is a crucial factor in the highly regulated financial industry. The results of this benchmark assessment also lay the groundwork for further studies in the two most important areas. To begin with, the integration of explainability tools such as SHAP or Anchors may help overcome the problem of transparency and establish user confidence in the use of automated decision-making tools [26]. Second, the issue of model fairness and bias mitigation methods must be thoroughly investigated so that the introduction of AI technologies in financial services is responsible, ethical, and inclusive. In summary, this study highlights the benefits of using complex machine learning algorithms in credit scoring, while also suggesting the need to consider both increased accuracy and accountability in high-stakes FinTech scenarios.

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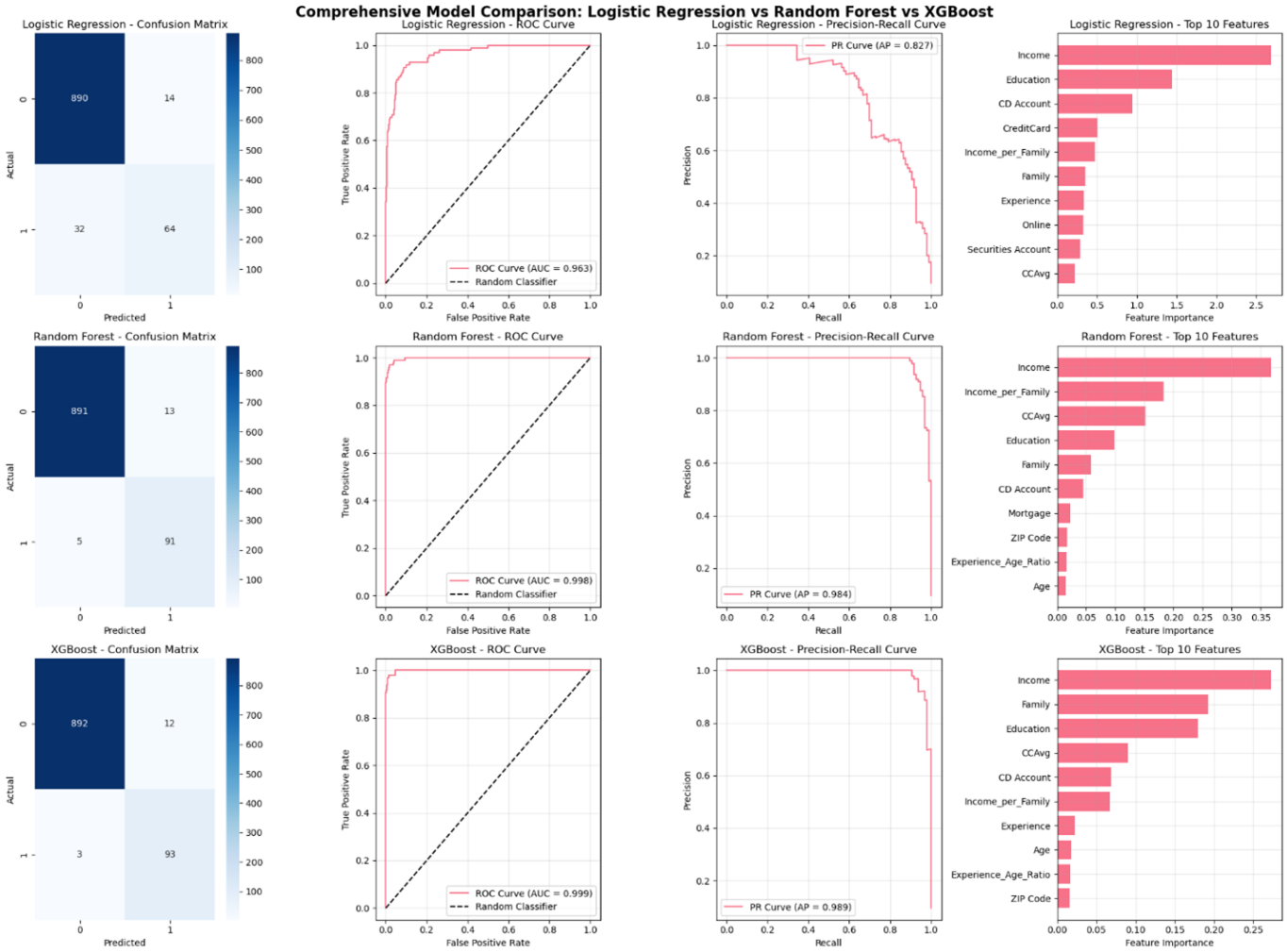

Fig. 1. Comprehensive model comparison: Logistic Regression, Random Forest, and XGBoost.

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